Proceedings of Autonomous Vehicle Laboratory
ATAV Technical Report Series
March 2026
March 4-25, 2026, Pomona, CA, USA

COMPARATIVE STUDY OF ACTIVE AND SEMI-ACTIVE SUSPENSION FOR AN ATAV PLATFORM: QUARTER-CAR VERTICAL DYNAMICS MODEL, ODE45 TIME-DOMAIN RESPONSE, AND RMS/TRAVEL METRICS

Prepared by:
John Vhen Ocsona

Autonomous Vehicle Laboratory
California State Polytechnic University, Pomona

Table Of Contents

1. Introduction……...……. 3
2. Methods……3
    - 2.1 Quarter-Car Model……...3
    - 2.2 Equations of Motion……...……. 3
    - 2.3 Control Strategies…….……... 3
        - 2.31 Active Control…… 3
        - 2.32 Semi-Active Control……. 4
    - 2.4 Road Inputs……... 4
    - 2.5 Simulation Setup and Metrics……..4
3. Data……...……4
    - 3.1 Figures…….…… 4
        - Figure 3.11……4
        - Figure 3.12……4
    - 3.2 Tables……...5
        - Table 3.21…….. 5
        - Table 3.22…….. 5
4. Results…….….5
    - 4.1 Step-Bump Results…….……. 5
    - 4.2 Washboard-Sinusoidal Results……...……. 5
5. Conclusion…….. 5

ABSTRACT

This report presents a quarter-car vertical dynamics simulation for a Baja-scale autonomous all-terrain vehicle (ATAV) to evaluate and compare passive, active, and semi-active suspension control strategies. The vehicle model consists of a sprung mass and an unsprung mass connected by a suspension spring-damper and a tire stiffness element. The system is simulated using MATLAB. Two road profiles are evaluated: an impulsive 2-inch step bump and a 5 Hz washboard sinusoidal input. Performance is quantified using RMS body acceleration for ride comfort and maximum suspension travel for packaging constraints and chassis stability. Results demonstrate that both controlled suspensions significantly improve dynamics over the passive baseline. While semi-active control yielded the lowest RMS acceleration on the washboard track (10.32 ft/s² vs. 10.99 ft/s² active), the fully active suspension demonstrated superior spatial performance and stability. On the impulsive step bump, the active system required significantly less maximum suspension travel than the semi-active system (0.191 ft vs. 0.223 ft) while maintaining excellent vibration isolation. These findings indicate that while semi-active control provides optimal pure ride comfort, the fully active suspension is the superior architecture for the ATAV platform, as it minimizes suspension working space and maximizes chassis stability within the vehicle's strict 9-inch design limit.

NOMENCLATURE

z_s	sprung mass displacement (ft)
z_u	unsprung mass displacement (ft)
$\dot{z}_s, \dot{z}_u$	velocities (ft/s)
$\ddot{z}_s$	body acceleration (ft/s²)
$r(t)$	road displacement input (ft)
m_s	sprung mass (slugs)
m_u	unsprung mass (slugs)
k_s	suspension stiffness (lbf/ft)
c_s	suspension damping (lbf·s/ft)
k_t	tire stiffness (lbf/ft)
F_a	active actuator force (lbf)
F_{max}	actuator saturation limit (lbf)
c_{min}, c_{max}	semi-active damping bounds (lbf·s/ft)

1. INTRODUCTION

Autonomous all-terrain vehicles (ATAV) require sophisticated suspension systems to balance chassis stability, sensor accuracy, and mechanical packaging within a highly constrained envelope. The fundamental goal of any suspension system is to isolate the vehicle body from road irregularities (ride quality) while maintaining firm contact between the tires and the ground (road holding). For an autonomous platform, minimizing chassis displacement and suspension travel is critical to ensure onboard sensors and cameras remain stable.

Traditional passive suspensions rely on fixed spring and damping coefficients, which limit the optimization for simultaneous comfort and handling, forcing designers to accept a compromised middle ground. To overcome this, variable suspension systems have been developed. Semi-active suspensions dynamically adjust the damping coefficient in real-time to dissipate energy, offering a low-power method to improve ride comfort. However, semi-active systems often require increased suspension-working space to achieve this comfort. Conversely, active suspensions inject external energy into the system via force actuators. Unlike a passive or semi-active damper, which can only dissipate energy, a force actuator can generate a stabilizing force in any direction regardless of the relative velocity across it.

This study utilizes a quarter-car model to investigate the vertical dynamics of passive, active, and semi-active suspension architectures, evaluating whether a fully active control strategy provides the optimal balance of vibration isolation and stability required for the ATAV platform [2].

2. METHODS

2.1 Quarter-Car Model

A quarter-car model represents one corner of the vehicle using two masses:

- **Sprung mass:** ¼ vehicle body supported at one wheel
- **Unsprung mass:** wheel/tire assembly mass

The suspension (spring and damper) connects the masses, and a tire stiffness connects the wheel to the road input $r(t)$.

Suspension travel:

$$x_{susp} = z_s - z_u$$

Tire deflection:

$$x_{tire} = z_u - r(t)$$

2.2 Equations of Motion

The vertical dynamics of the system are determined by applying Newton's second law to the sprung and unsprung masses [2]. The governing equations of motion [1][2] for the 2-DOF model are:

$$m_s\ddot{z}_s = -k_s(z_s - z_u) - c(z_s' - z_u') + F_a$$
$$m_u\ddot{z}_u = k_s(z_s - z_u) + c(z_s' - z_u') - k_t(z_u - r) - F_a$$

where c is either constant ($c = c_s$) or time-varying ($c = c_{eff}(t)$) for semi-active control.

2.3 Control Strategies

2.3.1 Active Control

Active control applies a force proportional to body velocity:

$$F_a = \text{sat}(-K_{act}\dot{z}_s, \mp F_{max})$$

This control opposes chassis vertical motion and includes a force limit representing actuator capability [1].

2.3.2 Semi-Active Control

Semi-active control modulates the damper coefficient only:

$$c_{eff}(t) \in [c_{min}, c_{max}]$$

A skyhook-style switching rule was implemented [3]:

- If $\dot{z}_s(\dot{z}_s - \dot{z}_u) > 0$, set $c_{eff} = c_{max}$
- otherwise set $c_{eff} = c_{min}$

Semi-active control does not inject net energy into the system; it changes energy dissipation via damping modulation.

2.4 Road Inputs

Two road profiles were evaluated:

1. **Step bump** (2 inches at $t = 1$s):

$$r(t) = 0.1667 \text{ ft} \cdot 1(t \geq 1)$$

2. **Washboard sine** (0.04 ft amplitude at 5 Hz)

$$r(t) = A\sin(2\pi f t) \qquad A = 0.04 \text{ ft}, f = 5 \text{ Hz}$$

2.5 Simulation Setup and Metrics

MATLAB ode45 was used for time-domain integration over 0–3 s with output step 0.001 s. Initial conditions were set to rest:

$$[z_s, \dot{z}_s, z_u, \dot{z}_u] = [0,0,0,0].$$

Performance metrics:

- **RMS body acceleration** computed over $t \geq 1$s:

$$a_{rms} = \sqrt{\frac{1}{T}\int a_s^2(t)\, dt}$$

- **Maximum suspension travel**:

$$x_{max} = \max \mid z_s - z_u \mid$$

3. DATA

3.1 Figures

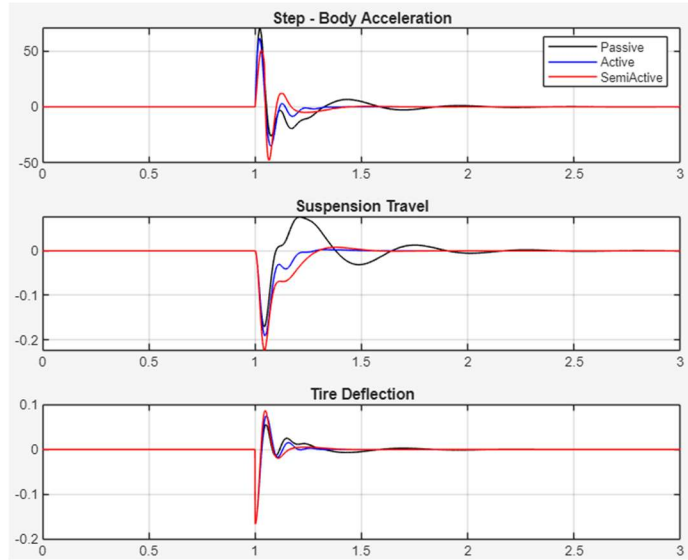

FIGURE 3.11: STEP-BUMP RESPONSE OVER TIME (ACTIVE VS. SEMI-ACTIVE VS. PASSIVE)

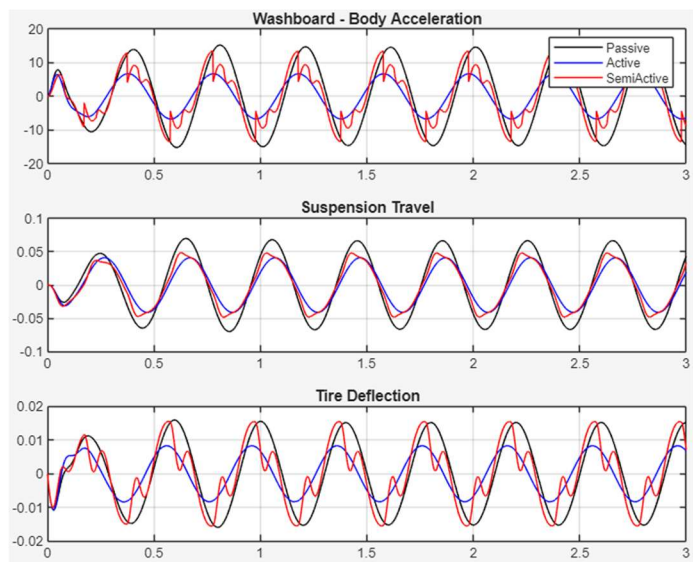

FIGURE 3.12: WASHBOARD RESPONSE OVER TIME (ACTIVE VS. SEMI-ACTIVE VS. PASSIVE)

3.2 Tables

Table 3.21		
Control	RMS Body Accel (ft/s^2)	Max Susp Travel (ft)
Active	8.0756	0.19106
Semi-active	8.0988	0.22376
Passive	9.5148	0.17048

TABLE 3.21: RMS BODY ACCELERATION AND MAX SUSPENSION TRAVEL DURING STEP-BUMP ROAD PROFILE

Table 3.22		
Control	RMS Body Accel (ft/s^2)	Max Susp Travel (ft)
Active	10.9961	0.05717
Semi-active	10.3243	0.05897
Passive	11.5964	0.05845

TABLE 3.22: RMS BODY ACCELERATION AND MAX SUSPENSION TRAVEL DURING WASHBOARD ROAD PROFILE

4. RESULTS

4.1 Step-Bump Results

As shown in Figure 3.11 and summarized in Table 3.21, both active and semi-active controllers successfully isolated the chassis under impulsive disturbances, significantly lowering the RMS body acceleration compared to the passive baseline (9.51 ft/s^2). However, achieving this enhanced isolation required an increase in maximum suspension working space for both controlled architectures. Crucially, the fully active suspension required significantly less travel (0.191 ft) than the semi-active system (0.223 ft) to achieve a nearly identical RMS acceleration (8.07 ft/s^2 vs 8.09 ft/s^2). This mirrors findings in the literature, which demonstrate that while both systems increase suspension working space over a passive setup to improve ride comfort, fully active systems require a substantially smaller increase in travel compared to semi-active systems (9% vs 17% increase, respectively) [1].

4.2 Sinusoidal-Washboard Results

The simulation data for the 5 Hz continuous washboard input, visualized in Figure 3.12 and detailed in Table 3.22, demonstrates that the semi-active Skyhook controller achieved a lower RMS body acceleration (10.32 ft/s^2) than the active system (10.99 ft/s^2). As noted in recent literature, semi-active systems are uniquely capable of matching or occasionally surpassing fully active systems in specific sprung mass acceleration reduction metrics [1][3]. However, acceleration (comfort) must be weighed against spatial stability (handling). The active system once again required the least amount of suspension travel (0.057 ft), outperforming both the semi-active and passive systems in road-holding metrics. Because the fully active suspension introduces active stabilizing energy into the system in the form of a force, it produces a less oscillatory response that returns quickly to equilibrium, whereas a semi-active suspension can only passively adjust its damping coefficient. This reinforces that active suspension is the optimal choice for prioritizing tight packaging constraints and chassis stability over pure passenger comfort.

5. CONCLUSION

This study demonstrates that implementing control strategies significantly improves the vertical dynamics of an ATAV platform compared to traditional passive suspensions. Through time-domain simulations of a quarter-car model, both active and semi-active models successfully reduced the RMS body acceleration of the chassis under impulsive and periodic road disturbances. While the semi-active Skyhook control achieved the lowest RMS body acceleration during continuous periodic excitation (10.32 ft/s^2), comfort must be weighed against spatial stability and road-holding.

Ultimately, the fully active suspension is the optimal choice for the ATAV platform. Active control provided superior chassis stability and required significantly less maximum suspension working space across the tested terrain profiles, specifically minimizing maximum travel to 0.191 ft compared to the semi-active system's 0.223 ft on the step bump. Because an active actuator injects stabilizing energy into the system, it reduces the inherent compromise between ride comfort and handling, generating a less oscillatory response that returns to equilibrium quickly. By minimizing suspension travel, the active system ensures the vehicle maintains optimal sensor stability while strictly adhering to the platform's compact mechanical packaging constraints.

APPENDIX

Appendix A: Model Parameters

$W_{total} = 400$ lbf
$W_{unsprung} = 20$ lbf per corner
$m_s = ((\frac{W_{total}}{4}) - W_{unsprung})/g$ slugs,
$m_u = W_{unsprung}/g$ slugs
$k_s = 6000$ lbf/ft
$c_s = 200$ lbf·s/ft
$k_t = 30000$ lbf/ft
Semi-active: $c_{min} = 50$, $c_{max} = 600$ lbf·s/ft
Active: K_{act}= 1000, $F_{max} = 2000$ lbf

Appendix B: MATLAB Implementation

```
%% setting up parameters
clear; clc; close all

% max actuator force
Fmax = 200;

%Gravity
g=32.2; %ft/s^2

%Weight assumptionsA
W_total = 400; %lbf
W_unsprung_per_corner = 20; %lbf

%Sprung weight corner
W_sprung_per_corner = (W_total/4)-
W_unsprung_per_corner;

%Convert to mass (slugs)
ms = W_sprung_per_corner/g;
mu = W_unsprung_per_corner/g;

% Suspension parameters (English units)
ks = 425;
cs = 22;
kt = 2500;

% Semi-active damping limits
c_min = 8;
c_max = 45;

% Step bump (2 inches)
r = @(t) 0.1667*(t >= 1);

% Active control gain
Kact = 35;

%% ODE Function Setup
function dx = quartercar_ode(t, x, ms, mu, ks, cs,
kt, r, Kact, mode, Fmax, c_min, c_max)

zs = x(1);
vs = x(2);
zu = x(3);
vu = x(4);

zr = (zs - zu);
vr = (vs - vu);

% Semi-active damping logic
if mode == 2
    if vs*vr > 0
        c_eff = c_max;
    else
        c_eff = c_min;
    end
else
    c_eff = cs;
end

fs = -ks * zr - c_eff .* vr;
ft = -kt*(zu - r(t));

% Active control logic
if mode == 1
    Fa_cmd = -Kact * vs;
    Fa = max(min(Fa_cmd, Fmax), -Fmax);
else
    Fa = 0;
end

as = (fs + Fa) / ms;
au = (ft - fs - Fa) / mu;

dx = [vs; as; vu; au];

end

%setup time vector for ode45
tspan = 0:0.001:3;
x0 = [0;0;0;0];

%% ================= STEP BUMP =================

% PASSIVE
mode = 0;
rhs = @(t,x) quartercar_ode(t, x, ms, mu, ks, cs,
kt, r, Kact, mode, Fmax, c_min, c_max);
[tP, xP] = ode45(rhs, tspan, x0);

% ACTIVE
mode = 1;
rhs = @(t,x) quartercar_ode(t, x, ms, mu, ks, cs,
kt, r, Kact, mode, Fmax, c_min, c_max);
[tA, xA] = ode45(rhs, tspan, x0);

% SEMI-ACTIVE
mode = 2;
rhs = @(t,x) quartercar_ode(t, x, ms, mu, ks, cs,
kt, r, Kact, mode, Fmax, c_min, c_max);
[tS, xS] = ode45(rhs, tspan, x0);

%% STEP OUTPUTS
```

```
[asP, suspP, tdP, rmsP, maxP] = compute_outputs(tP,
xP, ms, ks, cs, kt, r, Kact, Fmax, 0, c_min,
c_max);
[asA, suspA, tdA, rmsA, maxA] = compute_outputs(tA,
xA, ms, ks, cs, kt, r, Kact, Fmax, 1, c_min,
c_max);
[asS, suspS, tdS, rmsS, maxS] = compute_outputs(tS,
xS, ms, ks, cs, kt, r, Kact, Fmax, 2, c_min,
c_max);

figure('Name','Step Bump','NumberTitle','off')
subplot(3,1,1)
plot(tP, asP, 'k'); hold on; plot(tA, asA, 'b');
plot(tS, asS, 'r');
title("Step - Body Acceleration")
legend("Passive","Active","SemiActive"); grid on

subplot(3,1,2)
plot(tP, suspP, 'k'); hold on; plot(tA, suspA,
'b'); plot(tS, suspS, 'r');
title("Suspension Travel"); grid on

subplot(3,1,3)
plot(tP, tdP, 'k'); hold on; plot(tA, tdA, 'b');
plot(tS, tdS, 'r');
title("Tire Deflection"); grid on

fprintf("\n=== STEP BUMP ===\n")
disp(table([rmsP;rmsA;rmsS],[maxP;maxA;maxS],...

'VariableNames',{'RMS_ft_s2','MaxTravel_ft'},...
    'RowNames',{'Passive','Active','SemiActive'}))

%% ================= WASHBOARD =================

A = 0.04;
f = 5;
rwash = @(t) A * sin(2 * pi * f * t);

% PASSIVE
mode = 0;
rhs = @(t,x) quartercar_ode(t, x, ms, mu, ks, cs,
kt, rwash, Kact, mode, Fmax, c_min, c_max);
[tP, xP] = ode45(rhs, tspan, x0);

% ACTIVE
mode = 1;
rhs = @(t,x) quartercar_ode(t, x, ms, mu, ks, cs,
kt, rwash, Kact, mode, Fmax, c_min, c_max);
[tA, xA] = ode45(rhs, tspan, x0);

% SEMI-ACTIVE
mode = 2;
rhs = @(t,x) quartercar_ode(t, x, ms, mu, ks, cs,
kt, rwash, Kact, mode, Fmax, c_min, c_max);
[tS, xS] = ode45(rhs, tspan, x0);

%% WASH OUTPUTS
[asP, suspP, tdP, rmsP, maxP] = compute_outputs(tP,
xP, ms, ks, cs, kt, rwash, Kact, Fmax, 0, c_min,
c_max);
[asA, suspA, tdA, rmsA, maxA] = compute_outputs(tA,
xA, ms, ks, cs, kt, rwash, Kact, Fmax, 1, c_min,
c_max);
[asS, suspS, tdS, rmsS, maxS] = compute_outputs(tS,
xS, ms, ks, cs, kt, rwash, Kact, Fmax, 2, c_min,
c_max);

figure('Name','Washboard','NumberTitle','off')
subplot(3,1,1)
plot(tP, asP, 'k'); hold on; plot(tA, asA, 'b');
plot(tS, asS, 'r');
title("Washboard - Body Acceleration")
legend("Passive","Active","SemiActive"); grid on

subplot(3,1,2)
plot(tP, suspP, 'k'); hold on; plot(tA, suspA,
'b'); plot(tS, suspS, 'r');
title("Suspension Travel"); grid on

subplot(3,1,3)
plot(tP, tdP, 'k'); hold on; plot(tA, tdA, 'b');
plot(tS, tdS, 'r');
title("Tire Deflection"); grid on

fprintf("\n=== WASHBOARD ===\n")
disp(table([rmsP;rmsA;rmsS],[maxP;maxA;maxS],...

'VariableNames',{'RMS_ft_s2','MaxTravel_ft'},...
    'RowNames',{'Passive','Active','SemiActive'}))

%% ================= OUTPUT FUNCTION
=================
function [as, susp_travel, td, rms_acc, max_travel]
= compute_outputs(t, x, ms, ks, cs, kt, r, Kact,
Fmax, mode, c_min, c_max)

zs = x(:,1);
vs = x(:,2);
zu = x(:,3);
vu = x(:,4);

zr = zs - zu;
vr = vs - vu;

% Match semi-active damping in post-processing
if mode == 2
    c_eff = cs*ones(size(vs));
    for k = 1:length(vs)
        if vs(k)*vr(k) > 0
            c_eff(k) = c_max;
        else
            c_eff(k) = c_min;
        end
    end
else
    c_eff = cs*ones(size(vs));
end
```

```
fs = -ks * zr - c_eff .* vr;

if mode == 1
    Fa = -Kact * vs;
    Fa = max(min(Fa, Fmax), -Fmax);
else
    Fa = 0;
end

as = (fs+Fa)/ms;
susp_travel = zr;
td = zu - r(t);

idx = (t>=1);
rms_acc = sqrt(mean(as(idx).^2));
max_travel = max(abs(susp_travel));
end
```

REFERENCES

[1] Samaroo, K., Awan, A. W., Marimuthu, S., Iqbal, M. N., Daniel, K., and Shabbir, N., 2025, "Performance Investigation of Active, Semi-Active and Passive Suspension Using Quarter Car Model," *Algorithms*, **18**, 100.

[2] Jazar, R. N., 2008, *Vehicle Dynamics: Theory and Application*, Springer, New York, NY, pp. 741-742, 931.

[3] Mulla, A. A., Unaune, D. R., and Jalwadi, S. N., 2014, "Performance Analysis of Skyhook, Groundhook and Hybrid Control Strategies on Semiactive Suspension System," *Proc. International Conference on Automotive Technology (ICAT)*.